

Toxic cocktails in our soils : Assessing fungicide-insecticide mixture effects on earthworm growth and reproduction

Supplementary information

Lisa Gollot

Cleo Bodin

Laura Frattaroli

Rémy Beaudouin

Raphaël Royauté

Juliette Faburé

Table of contents

1	Earthworm culture	5
1.1	Breeding and experiment soil	5
1.2	DNA barcoding	6
2	Experiments	7
2.1	Growth and reproduction experiments	7
2.2	Experiment A - Single substance	7
2.2.1	Juvenile growth	7
2.2.2	Reproduction	11
2.3	Experiment B - Mixture	13
3	Jonker interaction models	17
3.1	Mathematical formulation	17
3.2	Computational implementation	18
3.3	Model comparisons	19
	References	21

List of Figures

1.1	Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow) .	6
2.1	Overview of the growth experiments	7
2.2	Overview of the reproduction experiments	7
2.3	Initial weight of juvenile earthworms.	8
2.4		9
2.5	Growth curves of juvenile earthworms.	10
2.6	Effect of the pesticides on juvenile growth rates	10
2.7	Initial weight of adult earthworms.	11
2.8	Effect of the pesticides on adult growth rates.	12
2.9	Experimental design of experiment B.	13
2.10	Expected effect range for each effect isobole (red points).	14
2.11	Initial body weight of earthworms used in the experiment B. Earthworm assigned to lot D are represented in blue while the ones assigned to lot E are represented in dark grey. . . .	14
2.12	Dose response curves obtained with cocoon production. The line represents the modelled dose-response curve (with common Y_{max} and slope for the 2 active substances). Crosses correspond to the number of cocoons produced by one cosm and the bold points represents the mean per dose.	15
2.13	Dose-response curves obtained for each mixture ratio. The lines represent the dose-response curves obtained (with common Y_{max} and Y_{min} set to 0). Crosses correspond to the number of cocoons produced by one cosm and the bold points represent the mean per condition. .	16
3.1	Results of the CA model fit on experimental data. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.	20
3.2	Results of the SA model fit on experimental data. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.	20

List of Tables

1.1	Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017). . .	5
2.1	NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method. . . .	11
2.2	Estimated parameters of the dose-response curve for cocoon production of earthworms exposed to Imidacloprid or epoxiconazole and their respective 95% confidence intervals. .	12
2.3	NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method. . . .	12
2.4	Estimated parameters of the dose-response curves from experiment A.	12
2.5	Estimated parameters of the dose-response curves from experiment A. Model with common slope.	13
2.6	ANOVA table	13
2.7	Estimated parameters of the dose-response curves from experiment B (common slope and Y_{max} , used in Jonker models.	15
3.1	Estimated interaction parameters by the different Jonker interaction models.	19
3.2	Comparison of the different models with Likelihood Ratio tests.	19

1 Earthworm culture

1.1 Breeding and experiment soil

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay ($< 2 \mu\text{m}$, g/kg)	226.0
Fine silt (2-20 μm , g/kg)	174.1
Coarse silt (20-50 μm , g/kg)	298.9
Fine sand (50-200 μm , g/kg)	239.1
Coarse sand (200-2000 μm , g/kg)	47.9
CaCO_3 total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

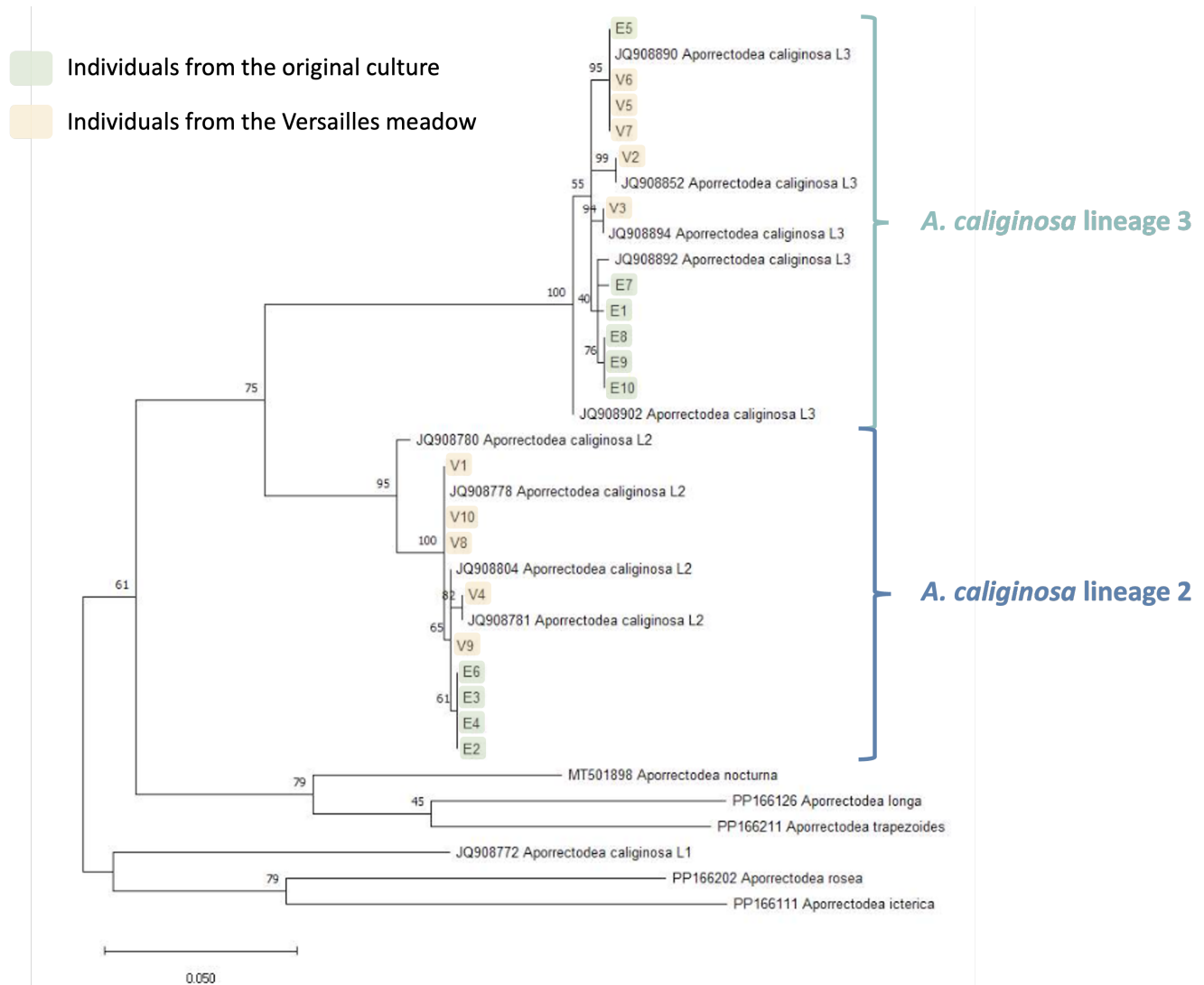


Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

2 Experiments

2.1 Growth and reproduction experiments



Figure 2.1: Overview of the growth experiments

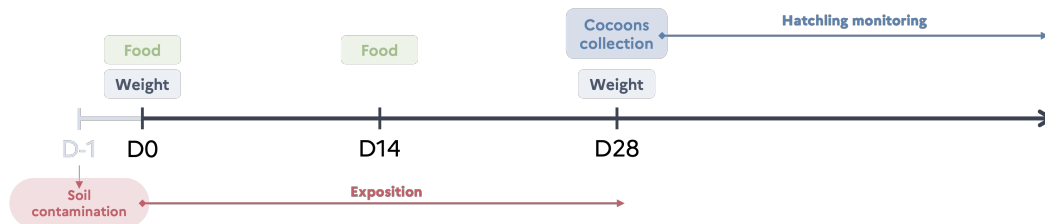


Figure 2.2: Overview of the reproduction experiments

2.2 Experiment A - Single substance

2.2.1 Juvenile growth

Initial state of earthworms

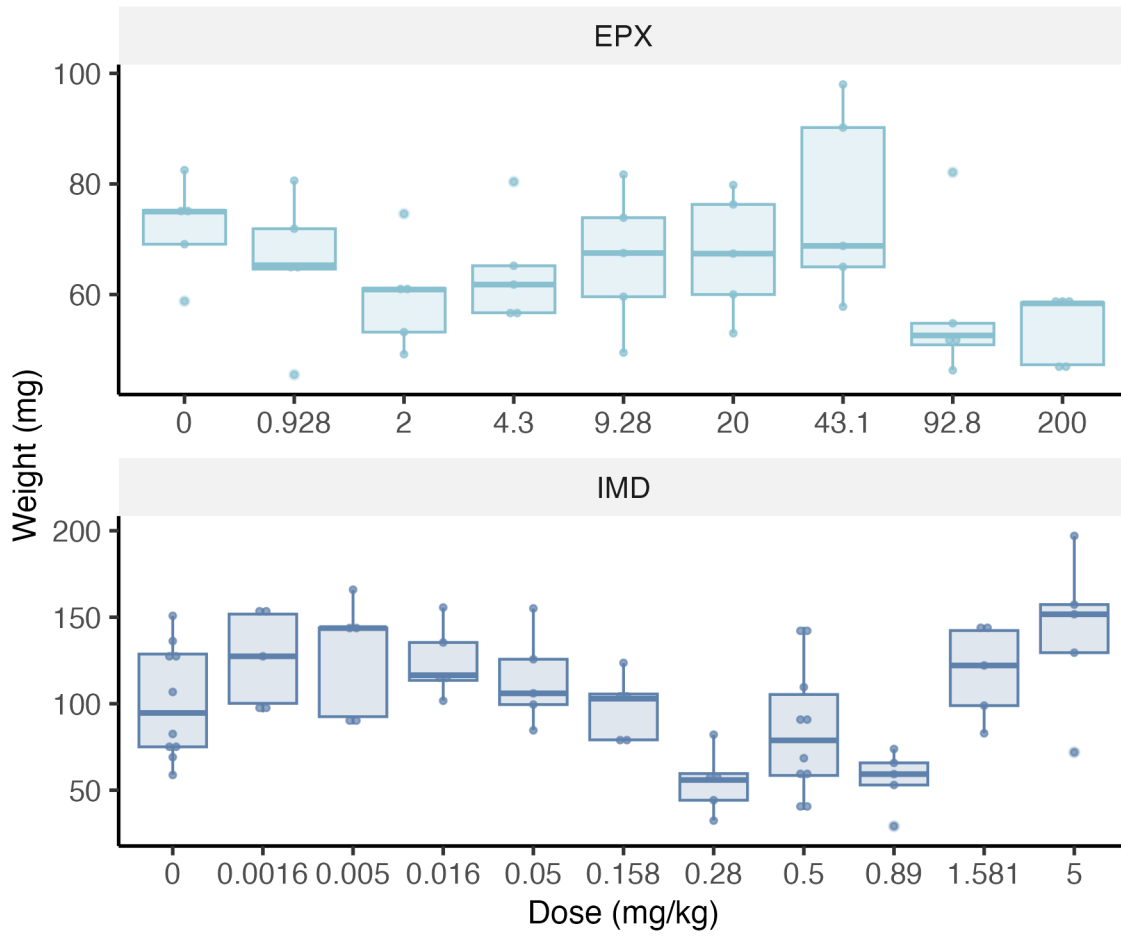


Figure 2.3: Initial weight of juvenile earthworms.

For IMD, there is a significant effect of the dose on earthworm initial weights.

- ANOVA : $pval(IMD) = 9.66e-05$,
- All concentrations are not significantly different from the control (post hoc Dunett test).

However, earthworms were non the less randomly attributed to conditions and the growth model allows to take the differences in initial weight into account.

For EPX, there is no significant effect of the dose on the initial weight of earthworms (Student test : $pval(IMD) = 0.125$).

Mortality during the experiment

Growth curves

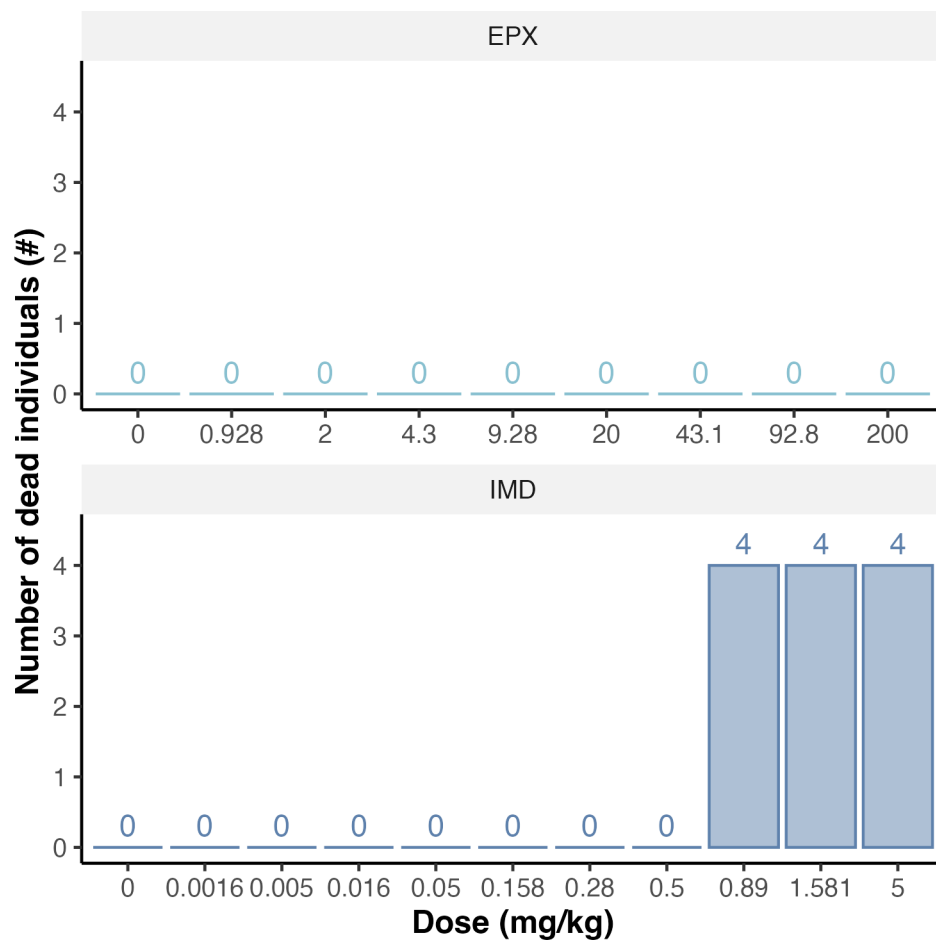


Figure 2.4

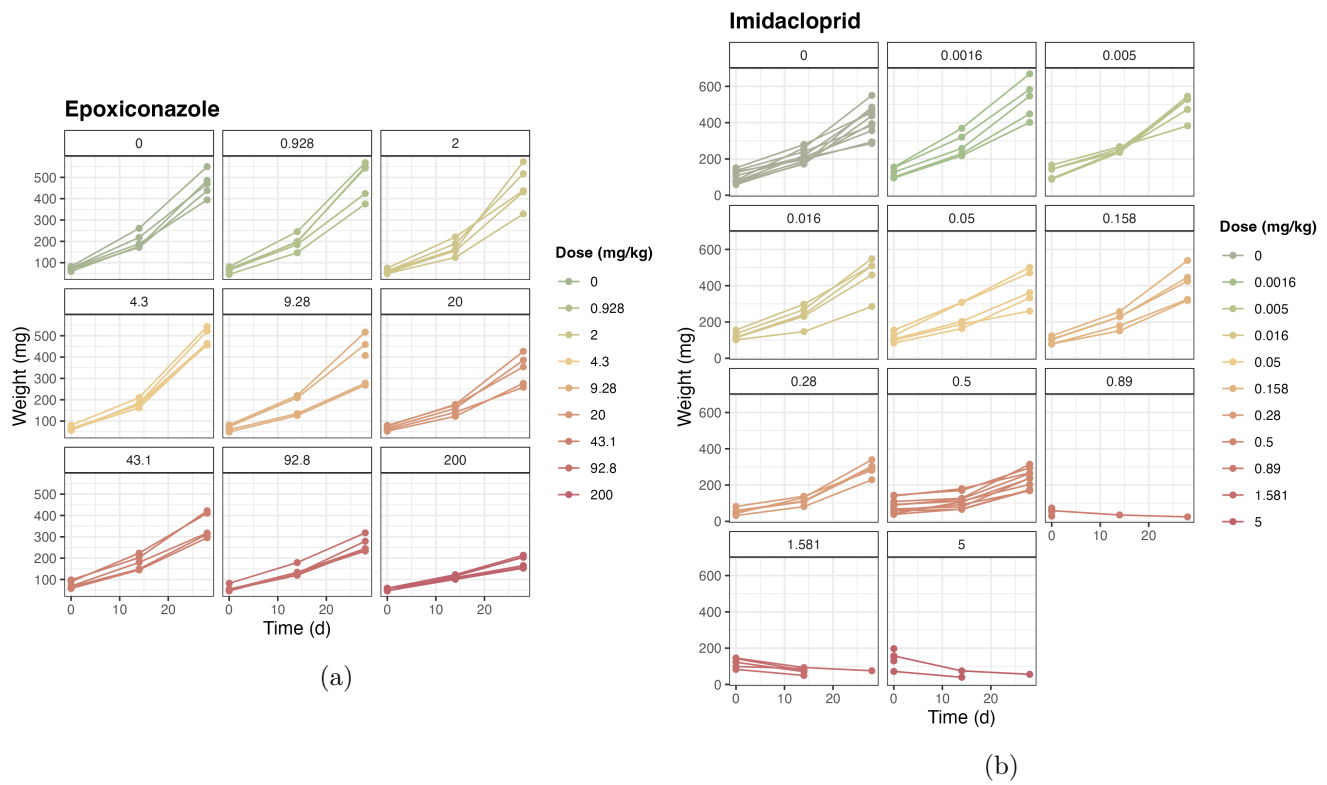


Figure 2.5: Growth curves of juvenile earthworms.

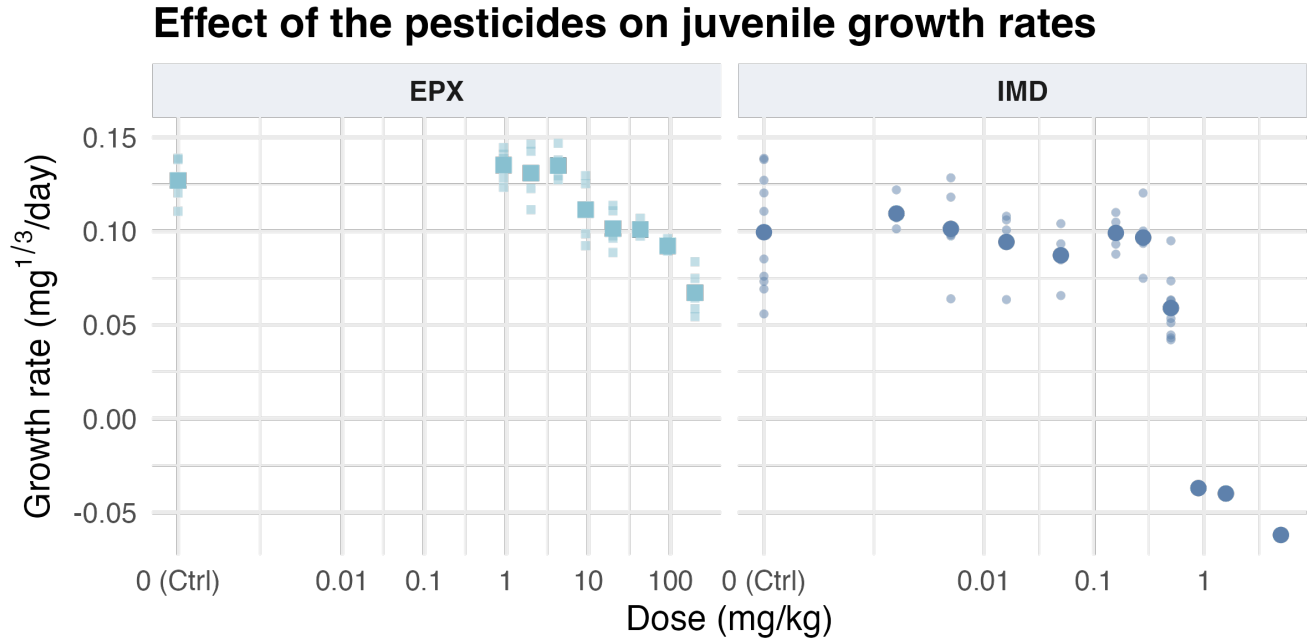


Figure 2.6: Effect of the pesticides on juvenile growth rates

Table 2.1: NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method.

	NOEC (mg/kg)	LOEC (mg/kg)
IMD	0.28	0.5
EPX	9.28	20

2.2.2 Reproduction

Initial state

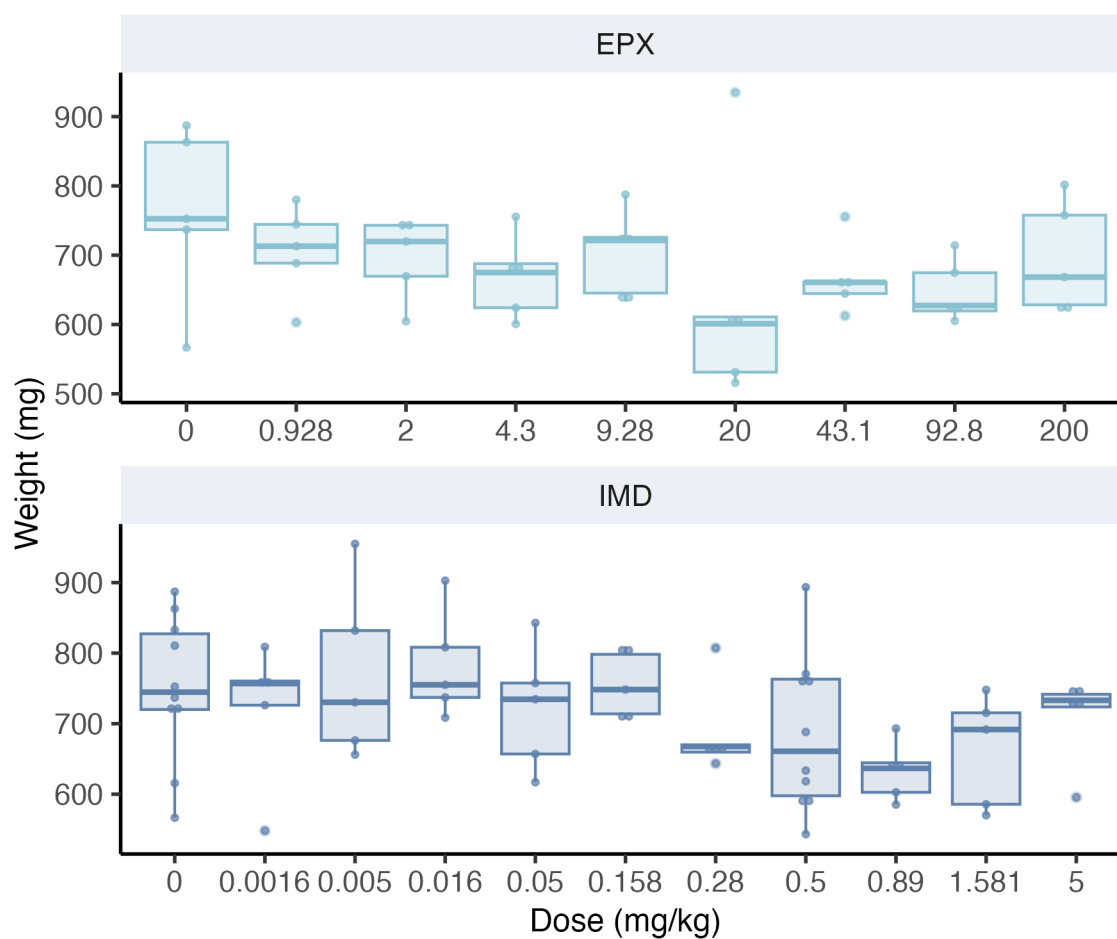


Figure 2.7: Initial weight of adult earthworms.

There is no effect of the dose on the initial weight of earthworms (Student test : $pval(EPX) = 0.17$ & $pval(IMD) = 0.573$).

Estimated effect

Table 2.2: Estimated parameters of the dose-response curve for cocoon production of earthworms exposed to Imidacloprid or epoxiconazole and their respective 95% confidence intervals.

	Imidacloprid	Epoxiconazole
Y_{min}	0 (fixed)	0 (fixed)
Y_{max} [CI 95%]	14.3 [13.0 ; 15.6]	18.7 [16.8 ; 20.7]
EC_{50} (mg/kg)	0.55 [0.45 ; 0.65]	126.8 [73.6 ; 180.0]
	126.8 [73.6 ; 180.0]	1.1 [0.5 ; 1.6]

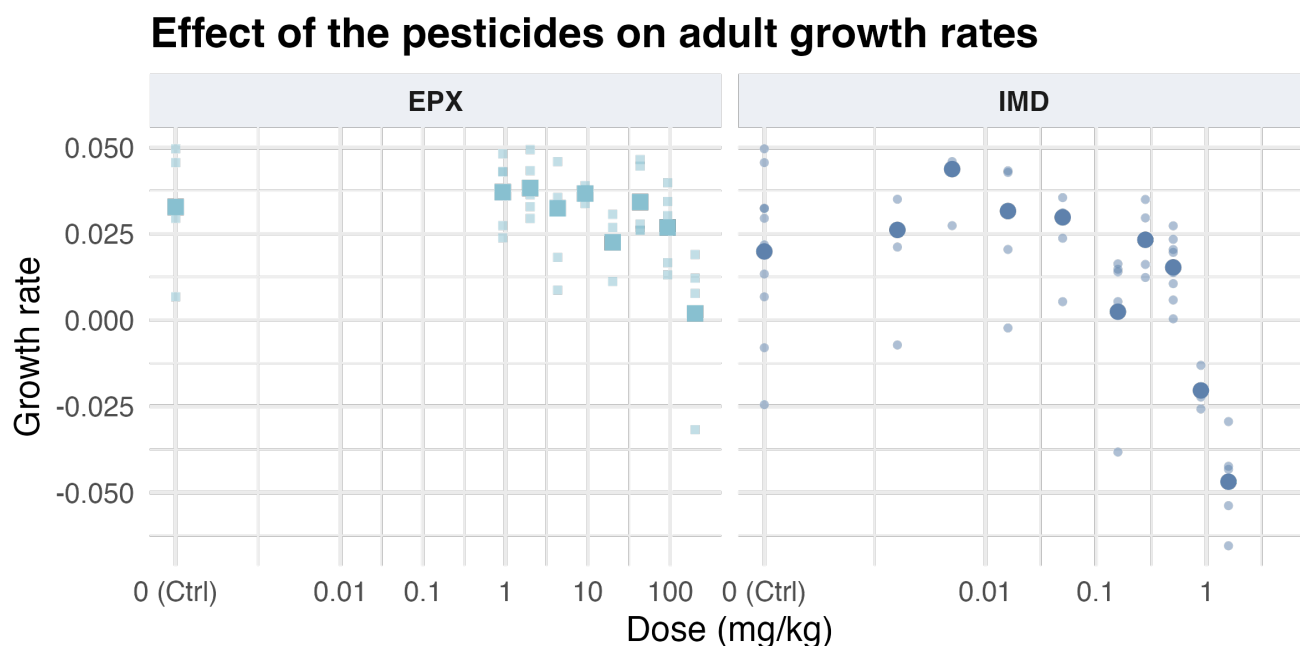


Figure 2.8: Effect of the pesticides on adult growth rates.

Table 2.3: NOEC and LOEC estimated with a t-test using the Bonferroni adjustment method.

	NOEC (mg/kg)	LOEC (mg/kg)
IMD	0.5	0.89
EPX	92.8	200

Table 2.4: Estimated parameters of the dose-response curves from experiment A.

	Y_{min} (#)	Y_{max} (#)	Slope	EC_{50} (mg/kg)
IMD	Set to 0	14.3 [13.0 ; 15.6]	2.8 [1.8 ; 3.7]	0.55 [0.45 ; 0.65]
EPX	Set to 0	18.7 [16.8 ; 20.7]	1.1 [0.5 ; 1.6]	126.8 [73.6 ; 180.0]

Table 2.5: Estimated parameters of the dose-response curves from experiment A. Model with common slope.

	Y_{min} (#)	Y_{max} (#)	Slope	EC_{50} (mg/kg)
IMD	Set to 0	14.3 [13.0 ; 15.6]	1.6 [0.6 ; 2.5]	0.55 [0.45 ; 0.65]
EPX	Set to 0	18.7 [16.8 ; 20.7]		126.8 [73.6 ; 180.0]

F-test - Comparison of the dose-response curve models with or without common slope with data normalized with respective controls :

Table 2.6: ANOVA table

Model	Model Df	RSS	Df	F value	p value
Different slopes	101	118059			
Common slope	100	117776	1	0.2404	0.6250

2.3 Experiment B - Mixture

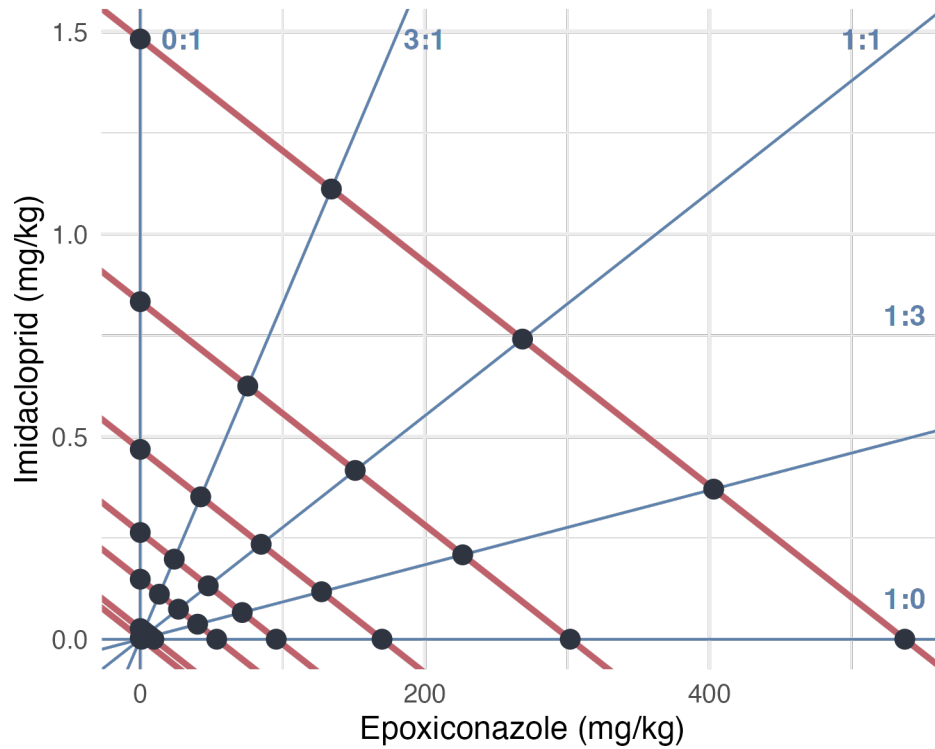


Figure 2.9: Experimental design of experiment B.

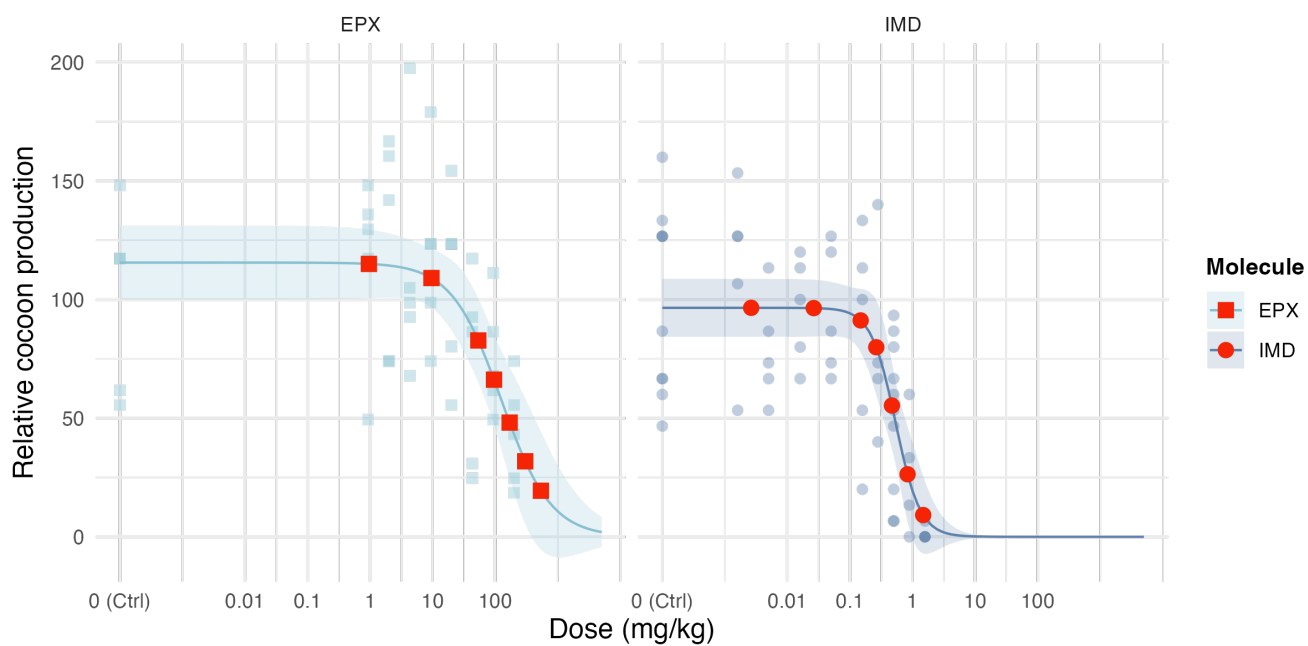


Figure 2.10: Expected effect range for each effect isobole (red points).

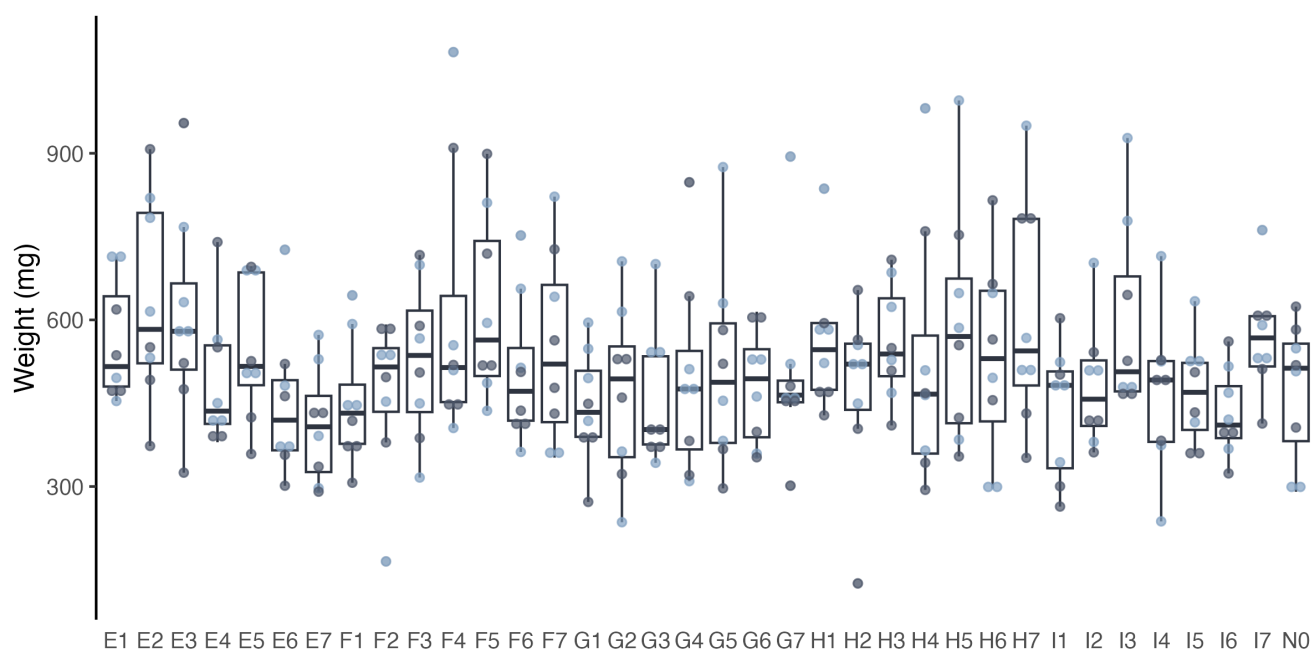


Figure 2.11: Initial body weight of earthworms used in the experiment B. Earthworm assigned to lot D are represented in blue while the ones assigned to lot E are represented in dark grey.

Table 2.7: Estimated parameters of the dose-response curves from experiment B (common slope and Y_{max} , used in Jonker models.

	Y_{min} (#)	Y_{max} (#)	Slope	$EC_{\{50\}}$ (mg/kg)
IMD	Set to 0	13.1 [11.6 ; 14.5]	2.3 [1.6 ; 2.9]	0.26 [0.20 ; 0.33]
EPX	Set to 0	13.1 [11.6 ; 14.5]	2.3 [1.6 ; 2.9]	215.6 [159.9 ; 271.2]

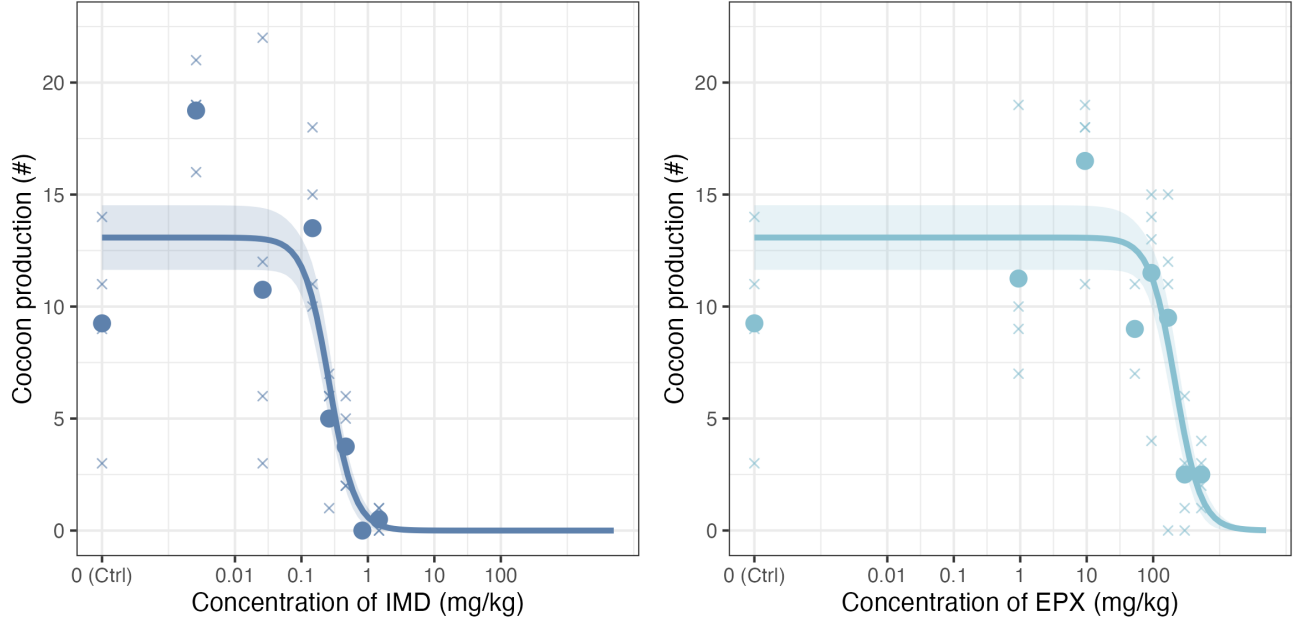


Figure 2.12: Dose response curves obtained with cocoon production. The line represents the modelled dose-response curve (with common Y_{max} and slope for the 2 active substances). Crosses correspond to the number of cocoons produced by one cosm and the bold points represents the mean per dose.

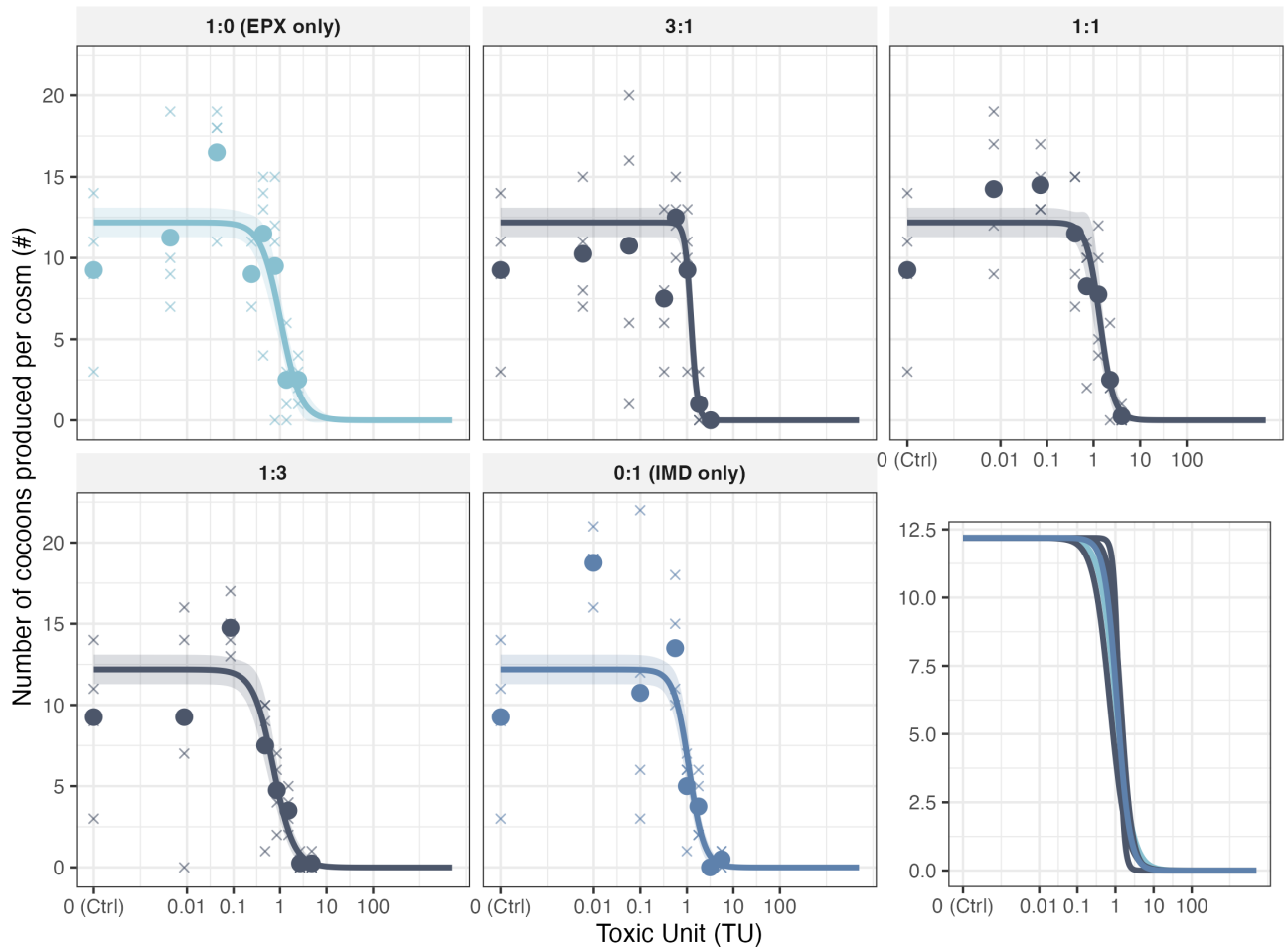


Figure 2.13: Dose-response curves obtained for each mixture ratio. The lines represent the dose-response curves obtained (with common Y_{max} and Y_{min} set to 0). Crosses correspond to the number of cocoons produced by one cosm and the bold points represent the mean per condition.

3 Jonker interaction models

3.1 Mathematical formulation

Let f_A and f_B be the functions describing the dose-response curves of substance A and substance B, respectively.

In the case of no interaction, concentration addition implies the dose-response surface of the mixture of A and B can be expressed as all couple of concentration of substance A and substance B ($c_A ; c_B$) as :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} = 1 \quad (3.1)$$

The Jonker simple interaction model add the definition of an interaction parameter (a) so the dose-response surface of the mixture of A and B can be expressed as all ($c_A ; c_B$) as :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} = \exp(G) \quad (3.2)$$

With :

$$G(z_A, z_B) = a \times (z_A \times z_B) \quad (3.3)$$

And :

$$z_i = \frac{\text{TU}_{x_i}}{\text{TU}_{x_A} + \text{TU}_{x_B}}, \quad \text{where } \text{TU}_{x_i} = \frac{c_i}{EC_{x_i}} \quad (3.4)$$

With :

- Y : The response for a given couple of concentrations,
- x : Usually 5, 10, ou 50 (50 used in this study),
- z_A & z_B : Contribution of each substance to the overall effect,

For the SA model, the interaction parameter value describes the type of the interaction :

- $a = 0$: Additivity,
- $a < 0$: Synergy,

- $a > 0$: Antagonisme.

The Jonker framework adds two other types of model, making it four models in total :

- CA : Concentration addition, case with no interaction (Equation 3.1)
- SA : Simple interaction with constant deviation from additivity (Equation 3.3),
- DL : Dose-level dependent interaction where effects vary with response magnitude (Equation 3.5),
- DR : Ratio-dependent interaction where effects vary with mixture composition (Equation 3.5).

The DL and DR models include a second interaction parameter, b . The expression of G is then :

$$G(z_A z_B) = \begin{cases} a [1 - b_{DL}] z_A z_B & \text{(DL model)} \\ (a + b_{DR} z_A) z_A z_B & \text{(DR model)} \end{cases} \quad (3.5)$$

3.2 Computational implementation

In practice, the fit of the dose-response curve surface is a double optimization :

1. For given set of interaction parameters, the corresponding effects must be found also numerically.
2. Numerical optimization of the interaction parameters

To perform step 1, a function `CA_complete2()` was written. For a given set of interaction parameters, finding the corresponding effect for a couple of concentration (c_A ; c_B) implies the minimization of the following term :

$$\frac{c_A}{f_A^{-1}(Y)} + \frac{c_B}{f_B^{-1}(Y)} - \exp(G) \quad (3.6)$$

For step 2, a function `CA_complete_fit_speed()` was written. Parameter estimation for the interaction models requires maximum likelihood estimation (MLE). The optimization procedure depends on the assumed error distribution: for normally distributed errors, this is equivalent to minimizing the residual sum of squares (calculated with the function `CA_complete2_RSS()`), while for Poisson-distributed errors, the log-likelihood function (Equation 3.7) (calculated with the function `CA_complete2_Poisson()`) is directly maximized.

Let $\mathbf{y} = (y_1, y_2, \dots, y_n)$ be the vector of observed responses, the parameter vector to be estimated, and $f(.|\cdot)$ the density function conditional to the parameters. Then, the likelihood function is defined as :

$$\mathcal{L}(\theta|\mathbf{y}) = \prod_{i=1}^n f(y_i|\theta)$$

The log-likelihood function, which is computationally more tractable, is given by:

$$l(\theta|\mathbf{y}) = \ln(\mathcal{L}(\theta|\mathbf{y})) = \sum_{i=1}^n \ln(f(y_i|\theta))$$

For count data (cocoon production), assuming $y_i \sim \mathcal{P}(\lambda_i)$ where $\lambda_i = g(\mathbf{x}_i; \theta)$ represents the expected response given covariates $\mathbf{x}_i$, the log-likelihood becomes:

$$\begin{aligned} l(\theta|\mathbf{y}) &= \sum_{i=1}^n \ln \left(\frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!} \right) \\ &= \sum_{i=1}^n [y_i \ln(\lambda_i) - \lambda_i - \ln(y_i!)] \\ &= \sum_{i=1}^n [y_i \ln(g(x_i; \theta)) - g(x_i; \theta) - \ln(y_i!)] \end{aligned} \tag{3.7}$$

3.3 Model comparisons

Table 3.1: Estimated interaction parameters by the different Jonker interaction models.

Model type	a	b	Log-Likelihood
CA	-	-	380.7
SA	-0.0045 [-2.5 ; 1.6]	-	380.7
DL	1.6 [-3.0 ; 14.0]	0.62 [-180.1 ; 0.71]	377.8
DR	-0.39 [-10.4 ; 4.7]	0.91 [-12.1 ; 31.7]	380.6

Table 3.2: Comparison of the different models with Likelihood Ratio tests.

	Log-Likelihood	Df	p-value
SA vs. CA	SA : 380.7 (CA : 380.7)	1	1
DL vs. CA	DL : 377.8	2	1
DR vs. CA	DR : 380.6	2	1

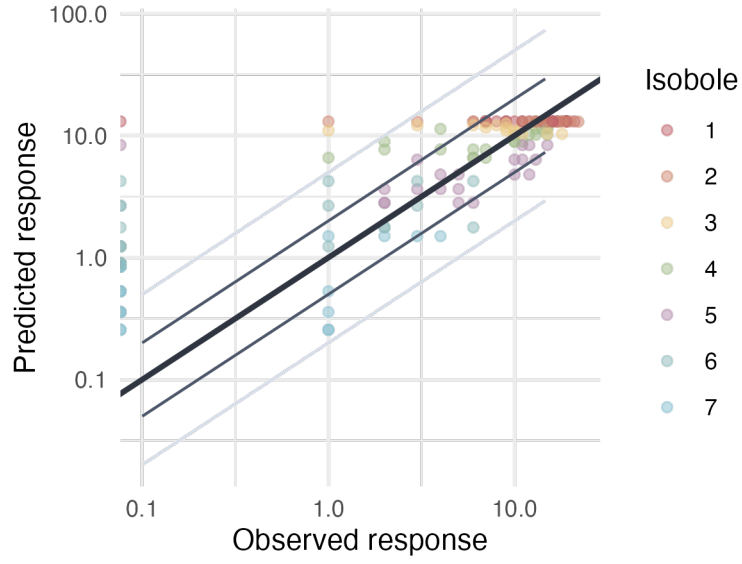


Figure 3.1: Results of the CA model fit on experimental data. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.

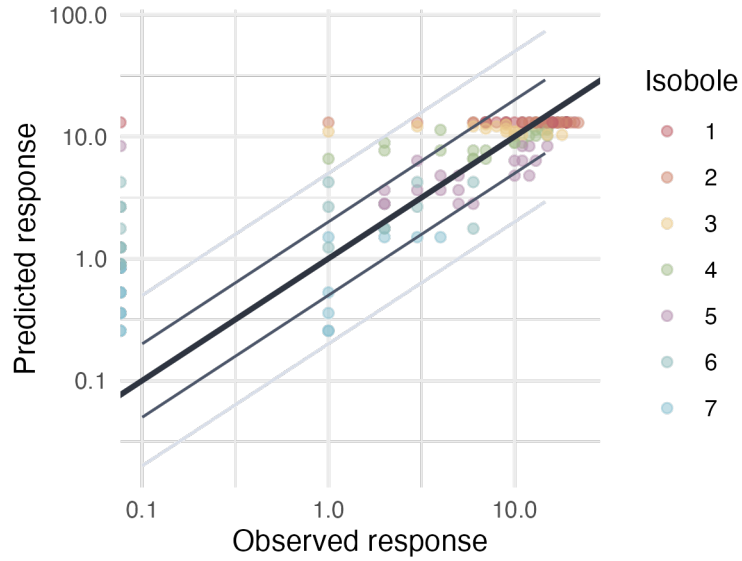


Figure 3.2: Results of the SA model fit on experimental data. Thin black and grey lines represent the 2-folds and 5-fold changes, respectively. The thicker black line represents the identity line.

Predicted responses will always be strictly above zero hence the points represented on the left hedge of the plot which correspond to the cosms where no cocoon were found.

References

Bart, S., Laurent, C., Péry, A. R. R., Mougin, C., & Pelosi, C. (2017). Differences in sensitivity between earthworms and enchytraeids exposed to two commercial fungicides. *Ecotoxicology and Environmental Safety*, 140, 177–184. <https://doi.org/10.1016/j.ecoenv.2017.02.052>